

Week 6: Data Visualisation and Dashboarding

Presented by:

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The Importance of Monitoring Supply Chain Performance

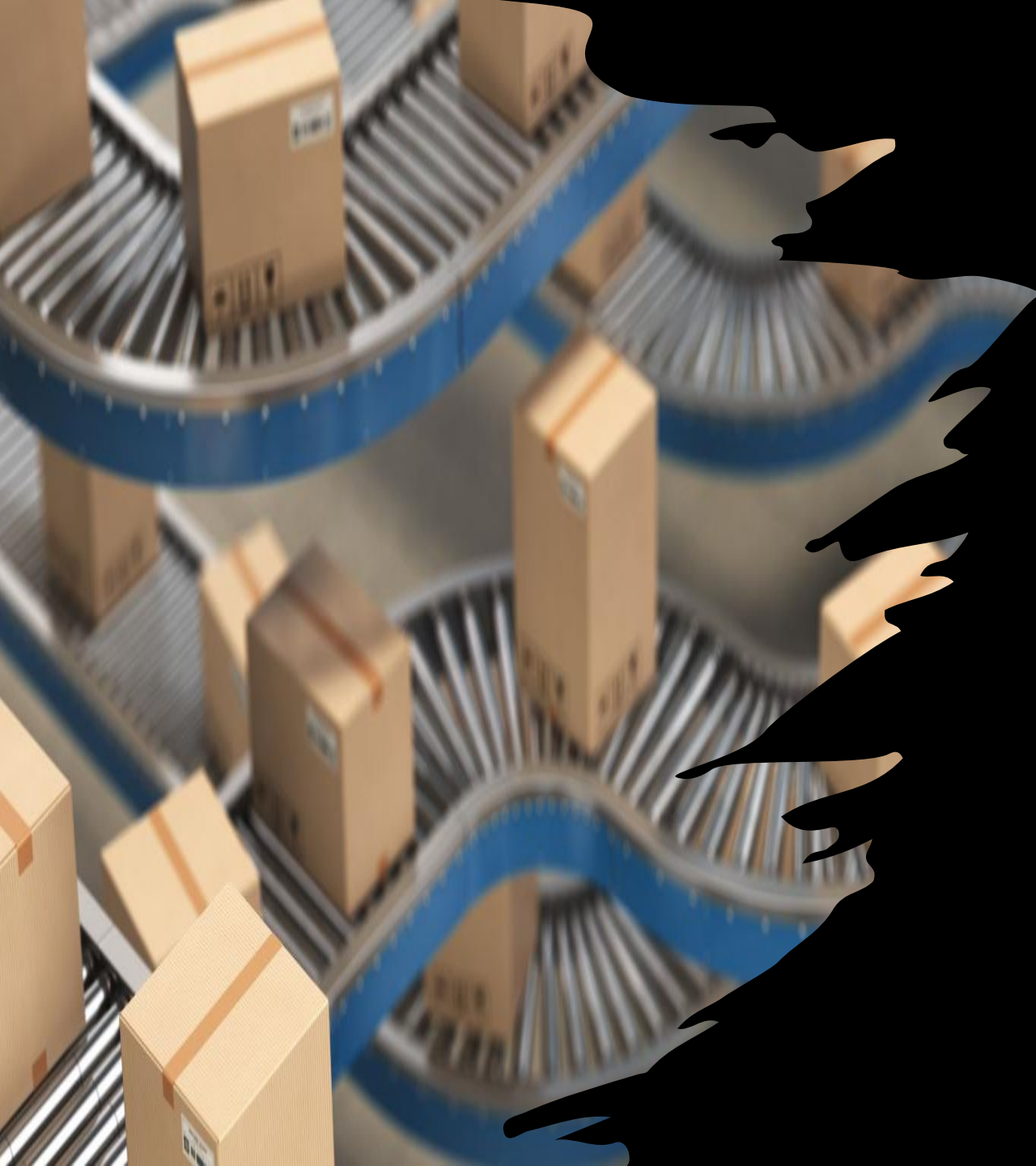
Cost Reduction: By identifying areas of waste and inefficiency, organisations can reduce costs.

Customer Satisfaction: Ensuring availability of inventory, on-time delivery and product quality

Risk Mitigation: Measuring performance helps identify and mitigate risks.

Supplier Relationships: Effective performance measurement enhances communication and collaboration with suppliers and enhances the chance of establishing long-term relationships with them.

This session, we are going to talk about KPIs and work to create a dashboard. A dashboard can be used as a tool to assist monitoring Supply Chain Performance and hence making decisions



Monitoring Supply Chain Performance: Key Performance Indicators (KPIs)



Key Performance Indicators (KPIs) are essential for both **supply chains** and **service chains** (often included in SLA)

KPIs' Examples in Service Chains:

Response Time

Delivery Time

Customer Complaint Rates

KPIs' Examples in Supply Chains:

Order Fulfillment Cycle Time: The time from order placement to delivery.

Inventory Turnover: How quickly inventory is sold or used.

Supplier Performance: Metrics related to supplier reliability, quality, and lead times.

Supply Chain Cost: Total cost of supply chain operations.

- What is the concept of service chains?
 - Provide some examples.
 - What is SLA?
 - Explain how monitoring dashboard of KPIs can help.
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- A horizontal white torn paper effect, resembling a piece of paper being pulled apart, runs across the middle of the slide, separating the list of bullet points from the rest of the page.

SERVICE CHAINS

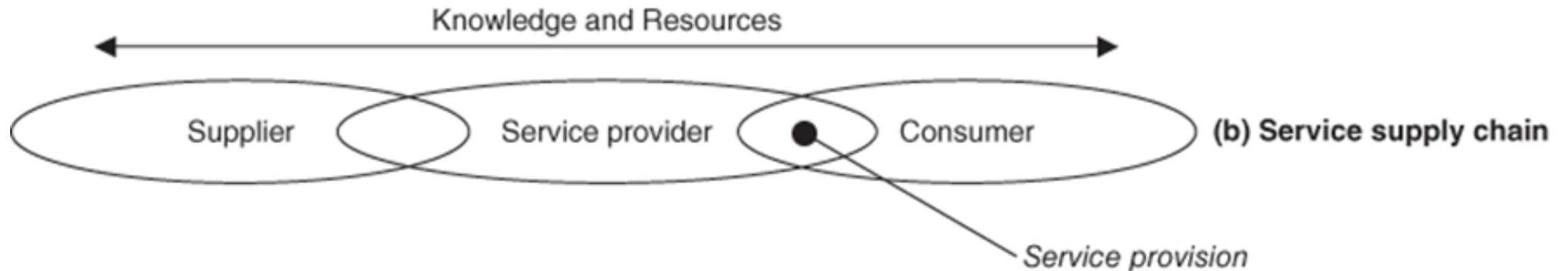
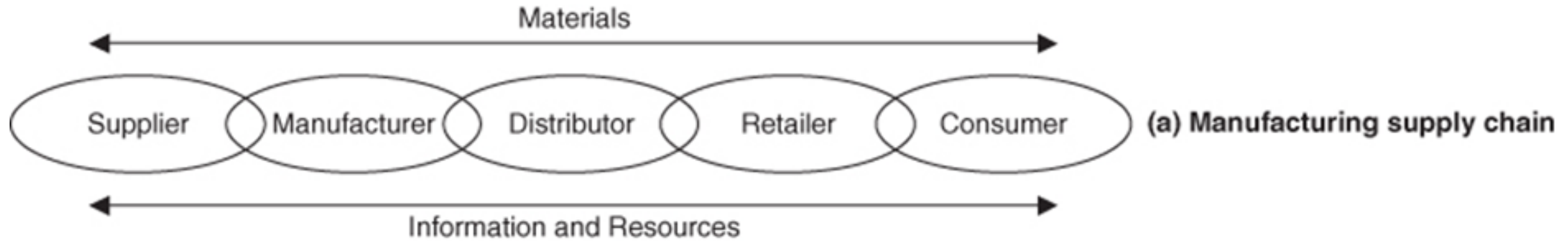
In SCM books and courses the focus is mainly **on supply chains in the manufacturing** context.

There is, however, a body of research that recognises the operations and SCM practices that are **required for the delivery of services.**

Internet Service Provider



Supply Chains vs Service Chains



KPIs need to have some sort of formulas to compute specific values

Perfect Order Measurement (POM):

- The percentage of orders that arrive complete, on time, damage-free, and with a correct invoice.
- $POM = (\%Complete) \times (\%On\ time) \times (\%Damage\ free) \times (\%Correctly\ invoiced)$

Cash to Cash Cycle Time (CCC):

- Measures the time it takes to convert inventory and accounts receivable into cash.
- $CCC = \text{Days Inventory Outstanding} + \text{Days Sales Outstanding} - \text{Days Payable Outstanding}$

Inventory Turnover (IT):

- Indicates how efficiently inventory is managed.
- $IT = \text{Cost of Goods Sold} / \text{Average Inventory}$

Lead Time:

- The time from placing an order to receiving it.
- $\text{Lead Time} = \text{Time of Order Placement to Receipt}$

Freight Cost per Unit Shipped:

- Measures shipping costs per unit.
 - $\text{Freight Cost per Unit Shipped} = \text{Total Freight Costs} / \text{Total Units Shipped}$
-

KPI Example: Inventory Turnover (IT): (the higher the ratio, the better)

Inventory turnover=

(cost of all goods sold)/(value of average inventory)

PROBLEM

The YouRace Company builds racing cars. In 2024, the **total cost of cars sold** was \$3 million. Its total inventory holding changed throughout the year, but **the average inventory holding** was worth \$250,000.

At the end of 2025, it implemented just-in-time principles to improve its inventory performance. In 2025, its sales increased and **the cost of cars was** \$4.5 million, while **the average inventory holding** was \$300,000.

Answer:

- Inventory turnover in 2024 = $3,000,000 / 250,000 = 12$
- Inventory turnover in 2025 = $4,500,000 / 300,000 = 15$



What is the best KPI for your company?

KPIs are defined differently based on what is important to an organisation and can therefore be different case to case.

Amazon:

- **Inventory Turnover**

Toyota:

- **Lead Time:** Measures the time from order placement to delivery.
- **Supplier Performance Metrics:** Evaluates suppliers based on **timely deliveries**.
- **Inventory Turnover:** Reflects efficient inventory management.



THE TRANSITION TO SERVICE ECONOMIES

Pre-industrial Era: Hand made and small-scale production

Industrial Revolution(1750, starting in Britain): Event encouraged countries to shift from agricultural economies to manufacturing economies. People moved off the land and into the mass production systems. This period can be termed as ‘post-agricultural’.

1. First Industrial Revolution (≈1750–1850): Steam engines, coal, textiles, iron, railways.
Shift from **handcraft** → **machine production**.

2. Second Industrial Revolution (≈1870–1914): Sometimes called the **Technological Revolution**.
Steel, electricity, petroleum, telegraph/telephone, light bulbs, internal combustion engines.
Rise of **mass production** and big industries like cars and chemicals.

3. Third Industrial Revolution (≈1970s onwards): Also called the **Digital Revolution**. Computers, electronics, telecommunications, internet. Shift from mechanical/electrical systems → **digital automation**.

4. Fourth Industrial Revolution (today): Term used by Klaus Schwab (World Economic Forum).
Technologies like **AI, robotics, IoT, 3D printing, biotechnology**.
Blurring boundaries between physical, digital, and biological world

In the second half of the 20th century, we observe the emergence of **services as industry**.

Rising **wealth and improved welfare** drive higher demand for value-added services.



Servitisation by Manufacturers

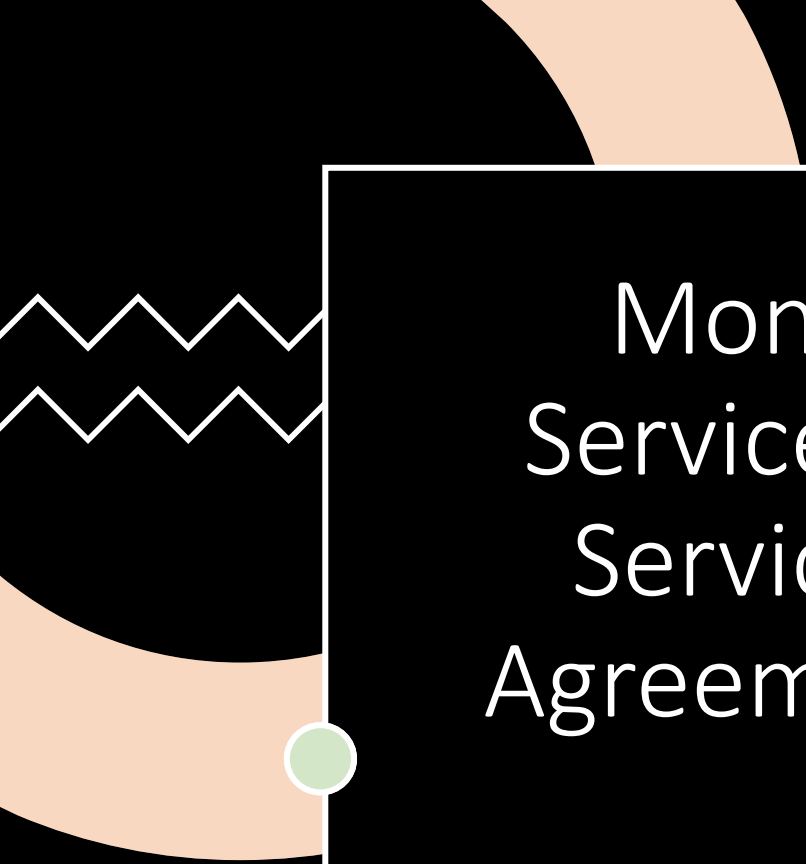
Manufacturers not are just delivering tangible, physical products but also offering services which enhance their value proposition and of course add to profitability.

Many manufacturers now offer finance packages to go with their products (e.g. most car manufacturers now also offer finance to their customers) as well as extended warranties, repair services, remote performance monitoring of the product (in trucking, for example) and so forth.

Rolls-Royce, for instance, earns around 50% of its revenue from services.



- 
- **Extended warranties**
e.g. dealership extended insurance
 - **Pet insurance**
 - **Funeral insurance**
 - **Flight cancellation insurance**
 - **Rental car insurance**
 - **Anything you buy, and they offer you insurance or warranty at some extra costs** (e.g. from a fridge, TV or sofa to a phone to a watch)
 - **Identity theft insurance**
 - **Shipping protection insurance**
 - **Accidental damage insurance**
 - **Event ticket insurance**
 - **Small appliance insurance**
 - **Dental insurance**
 - **Most extras in your car, home, health, car, or any other insurance policies**
- 



Monitoring Service Chains: Service Level Agreement (SLA)



Service Level Agreement (SLA)

- An SLA (service level agreement) outlines your expectations from the service provider (in terms of quality, availability, responsibilities, and penalties).

Service Level Agreement (SLA) is particularly important when the service is purchased or outsourced

Examples:

A call centre or IT, SLA can include:

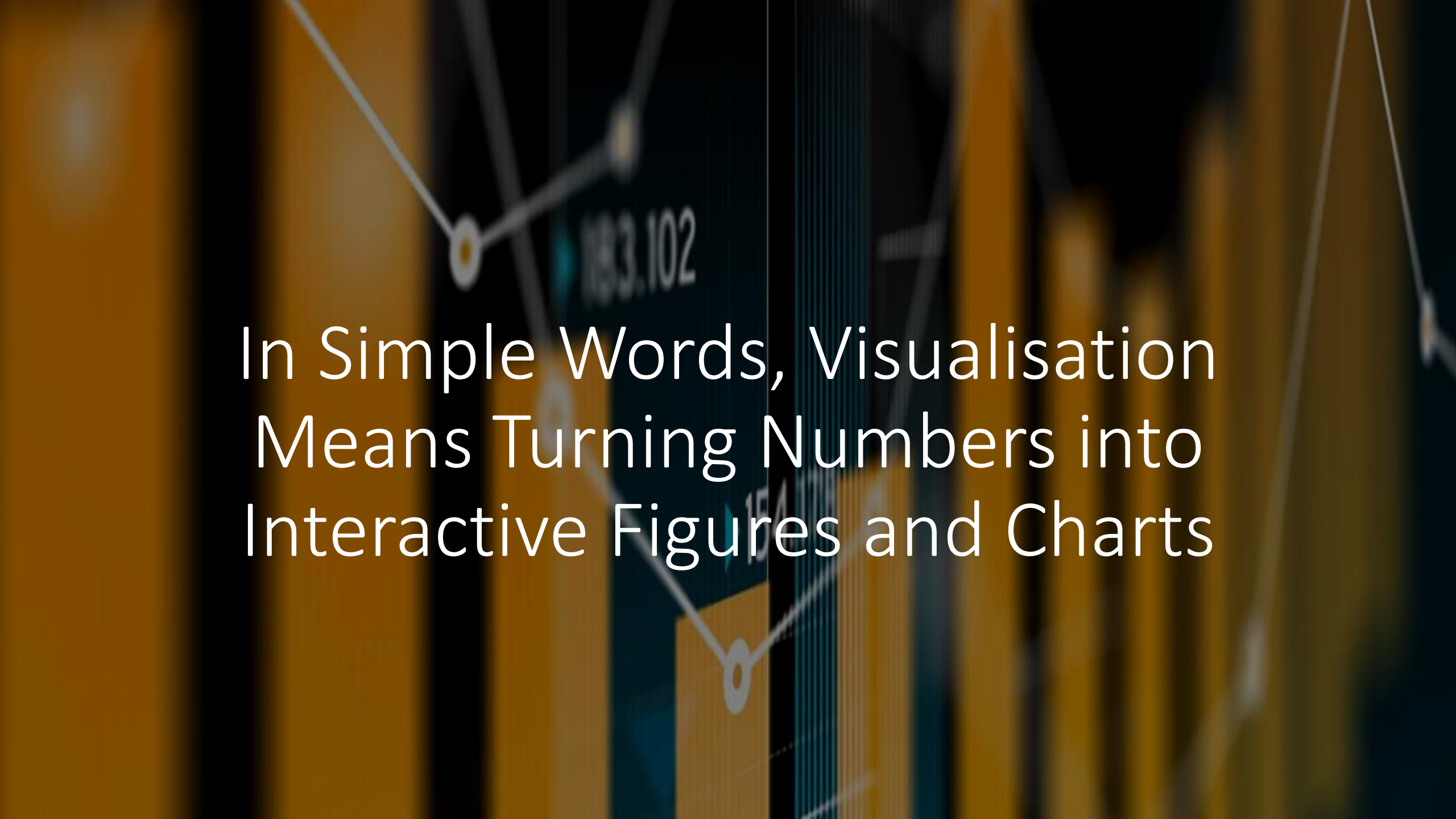
- **Availability Time**
- **Response Time**
- **Resolution Time**

Logistics and Transportation Services:

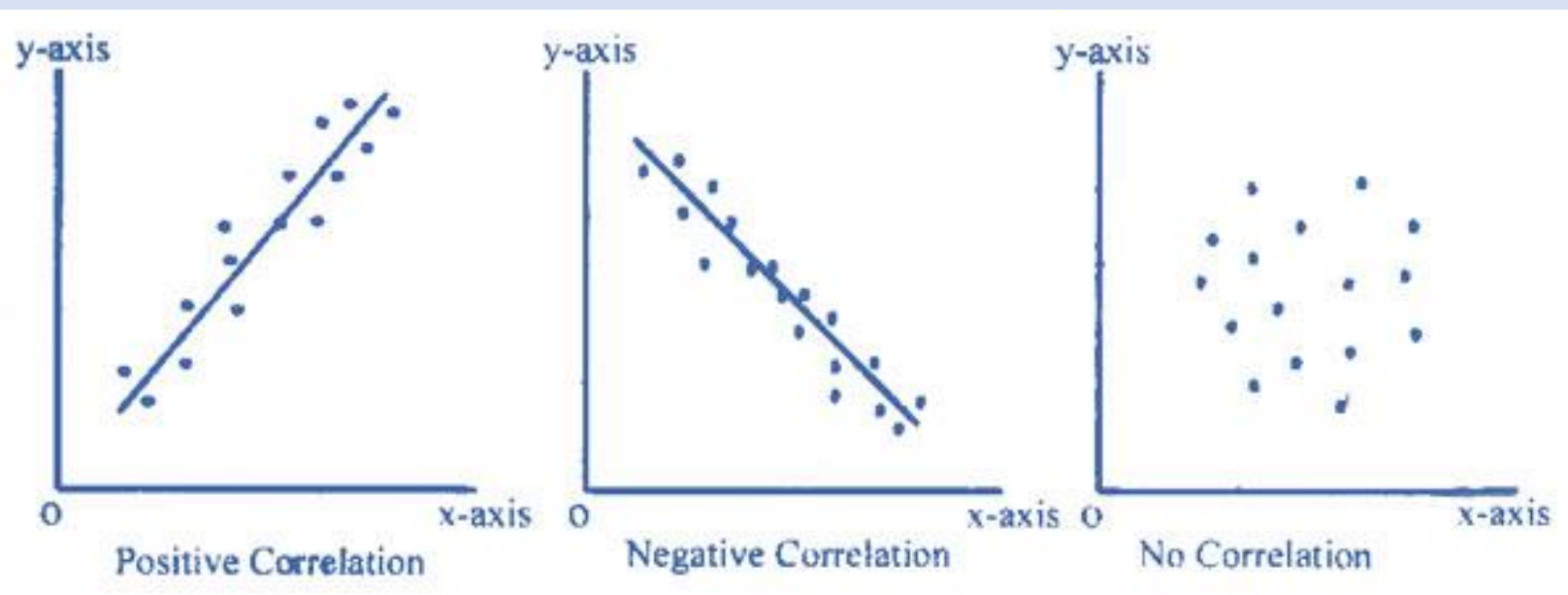
- **Delivery Time**
- **Allowed Damage Rate**
- **Penalty for delays or damages**

Visual Aid Tools and Dashboards: Monitoring KPIs and the Performance of Supply/Service Chains



The background is a blurred image of financial data visualizations. It features a line graph with a white line and circular markers on a dark blue background. A specific data point is labeled with the number '183.102' in a light blue font. Below this, another label '154.17' is partially visible. The overall aesthetic is professional and tech-oriented, with a color palette of dark blues, light blues, and warm oranges/browns.

In Simple Words, Visualisation
Means Turning Numbers into
Interactive Figures and Charts



Where visualisation is used



Changes over time : This is perhaps the most basic and common use of data visualisation, but that doesn't mean it's not valuable.



Determining frequency: Frequency is also a fairly basic use of data visualisation because it also applies to data that involves time. If time is involved, it is logical that you should determine how often the relevant events happen over time.



Determining relationships (correlations): Identifying correlations is an extremely valuable use of data visualisation. It is extremely difficult to determine the relationship between two variables without a visualisation, yet it is important to be aware of relationships in data.



Monitoring the supply chain performance: Monitoring performance of the supply chain, where there are daily data coming from a range of suppliers can be quite challenging. This becomes much easier by looking at visualised data.



Analysing value and risk: Determining complex metrics such as value and risk requires many different variables to be factored in, making it almost impossible to see accurately with a plain spreadsheet. Data visualisation can be as simple as color-coding a formula to show which opportunities are valuable and which are risky.

Data visualisation should be...

Informative

Data visualisation should be able to convey the desired **information from data to the reader.**

Efficient

Data visualisation should **not be ambiguous.**

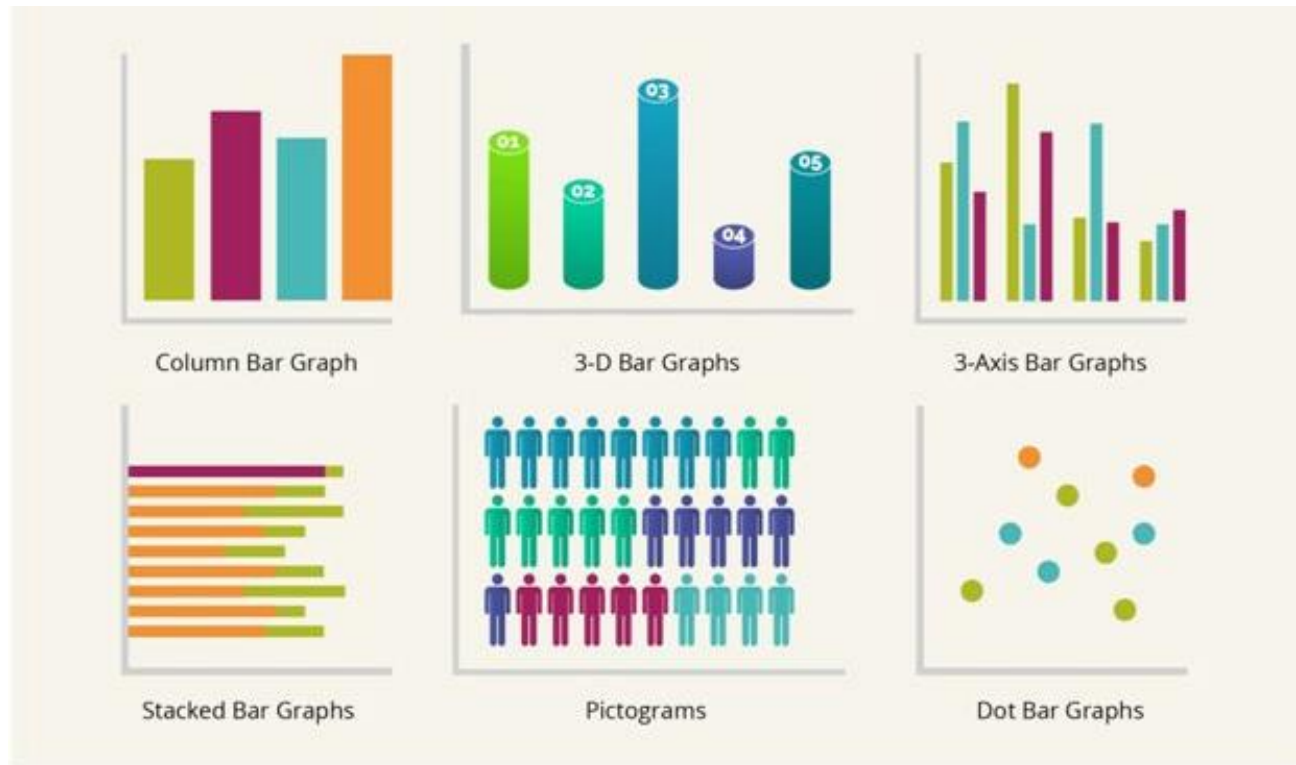
Appealing

Data visualisation should be captivating and **visually pleasing.**

Interactive and Predictive (Optional)

Data visualisations can contain **variables and filters with which the users may interact** to predict results of different scenarios.

Different types of graphs



Column /Bar charts are used to present separate values, especially the differences between these values. They can be used for comparison purpose.

- **Advantages:** summarise a large dataset in visual form; easily compare two or more data sets; better clarify trends than do tables; estimate key values at a glance.
- **Variations:** 3-D Bar Graphs; 3-Axis Bar Graphs; Stacked Bar Graphs ; Line/Dot Bar Graphs; Histograms; Pictograms.

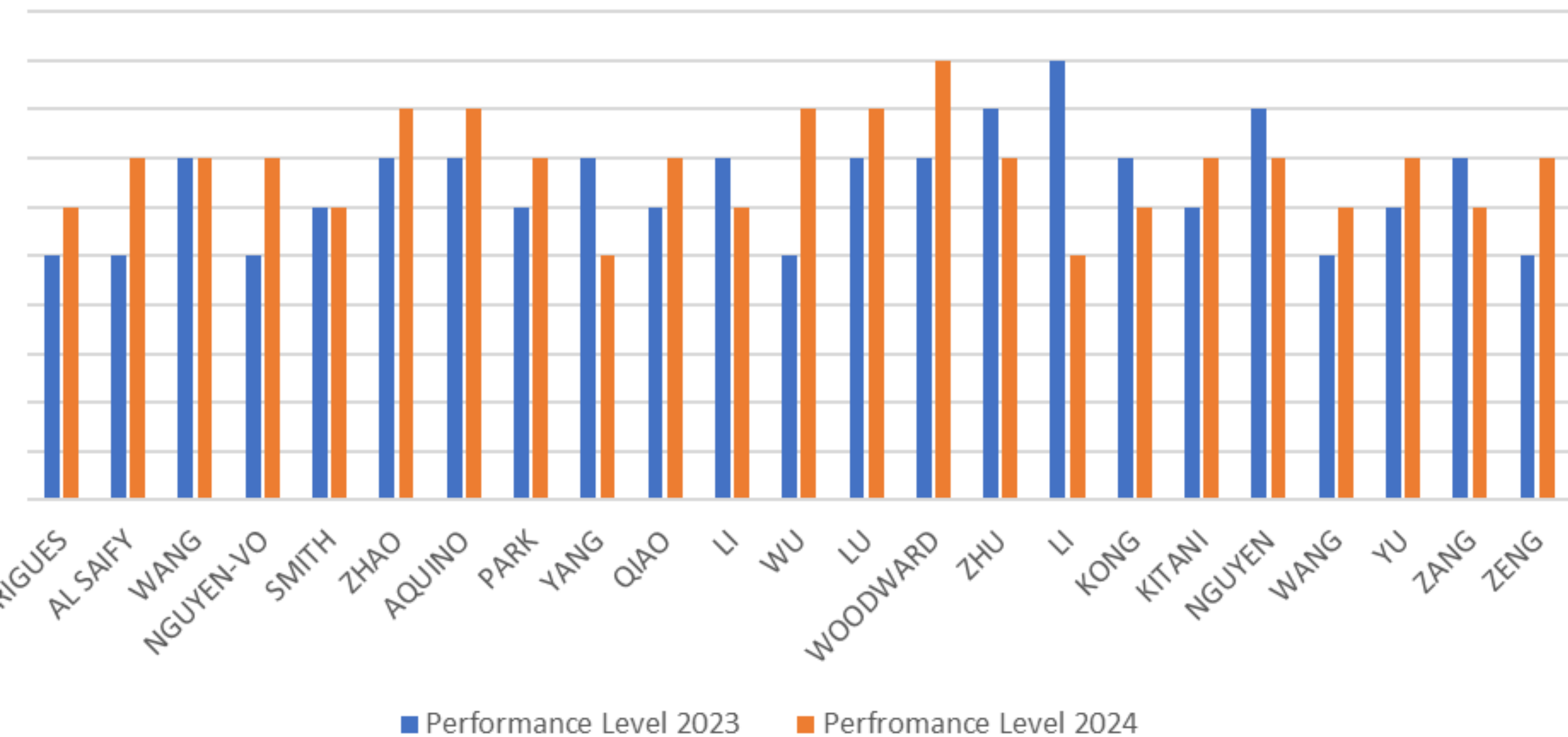


In a meeting with CEO, you become in charge of improving the performance of the production lines of the company.

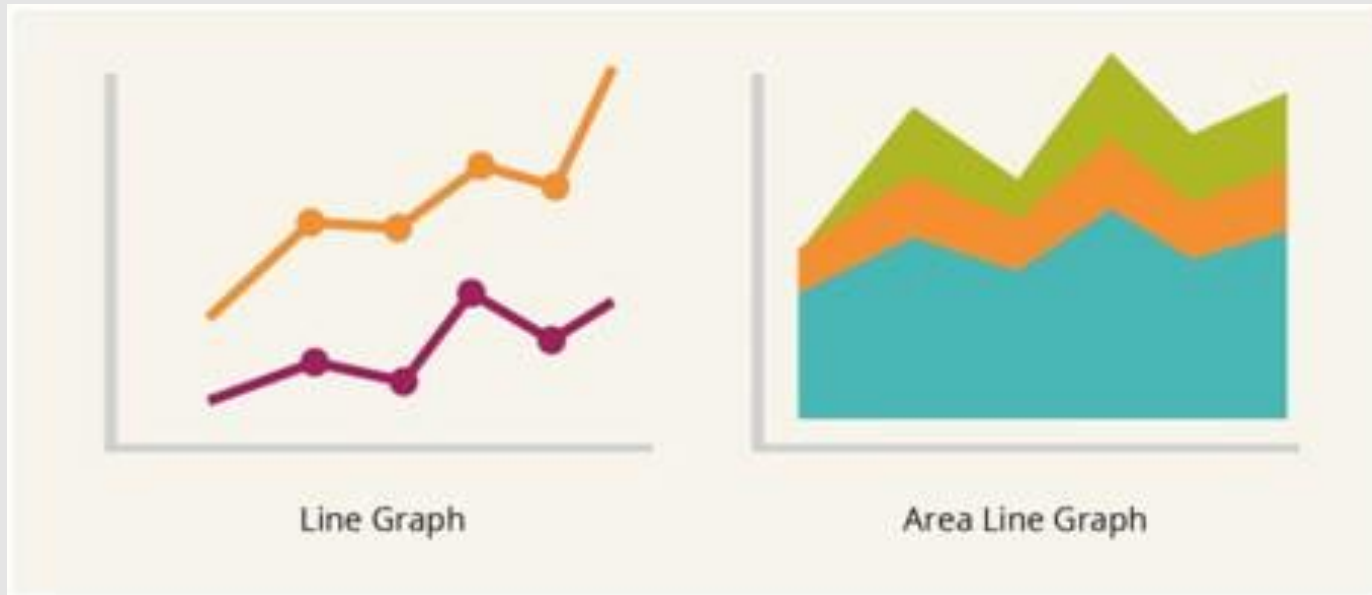
Let's evaluate the performance of production line 1



Chart Title



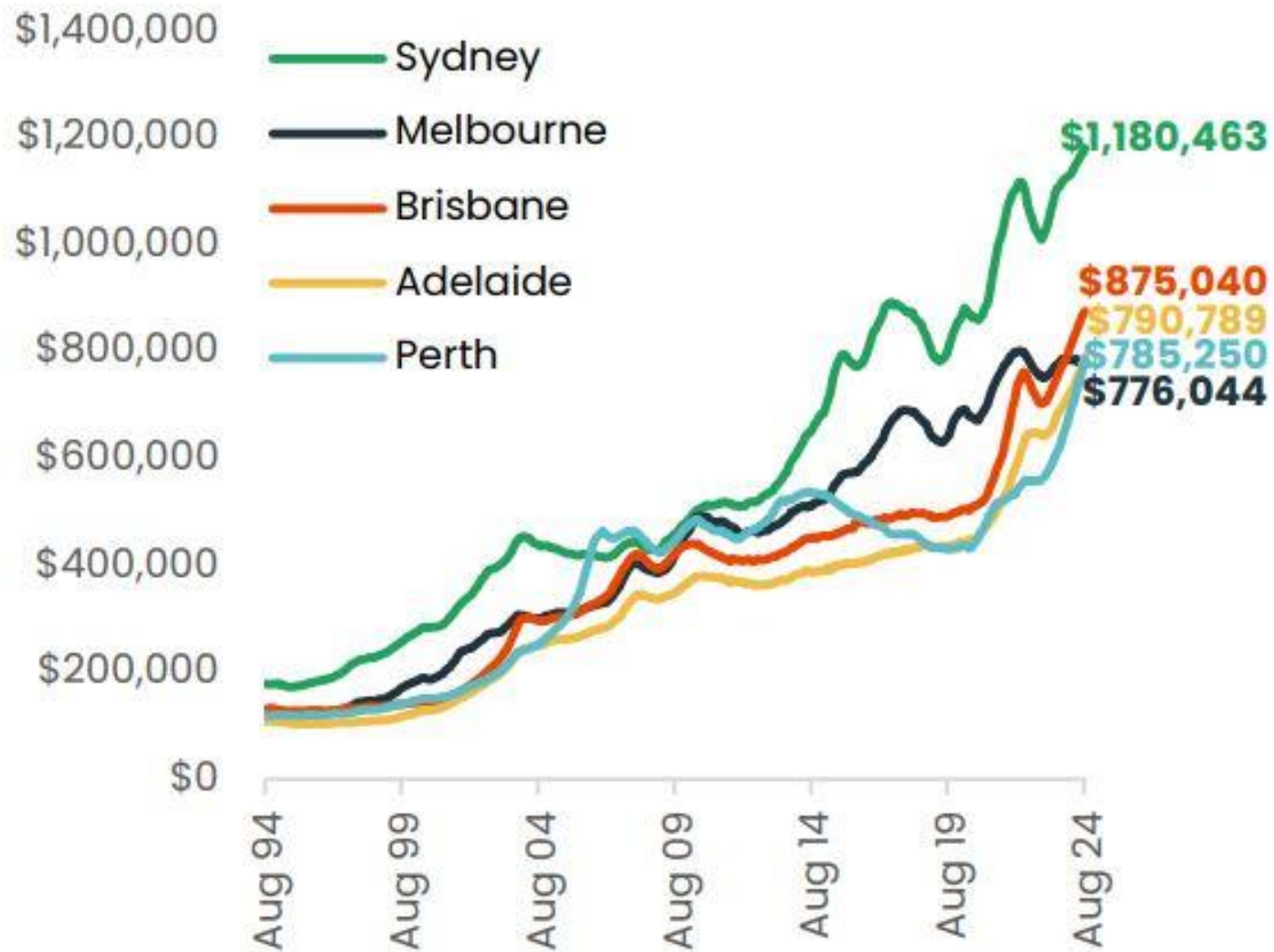
Line Charts

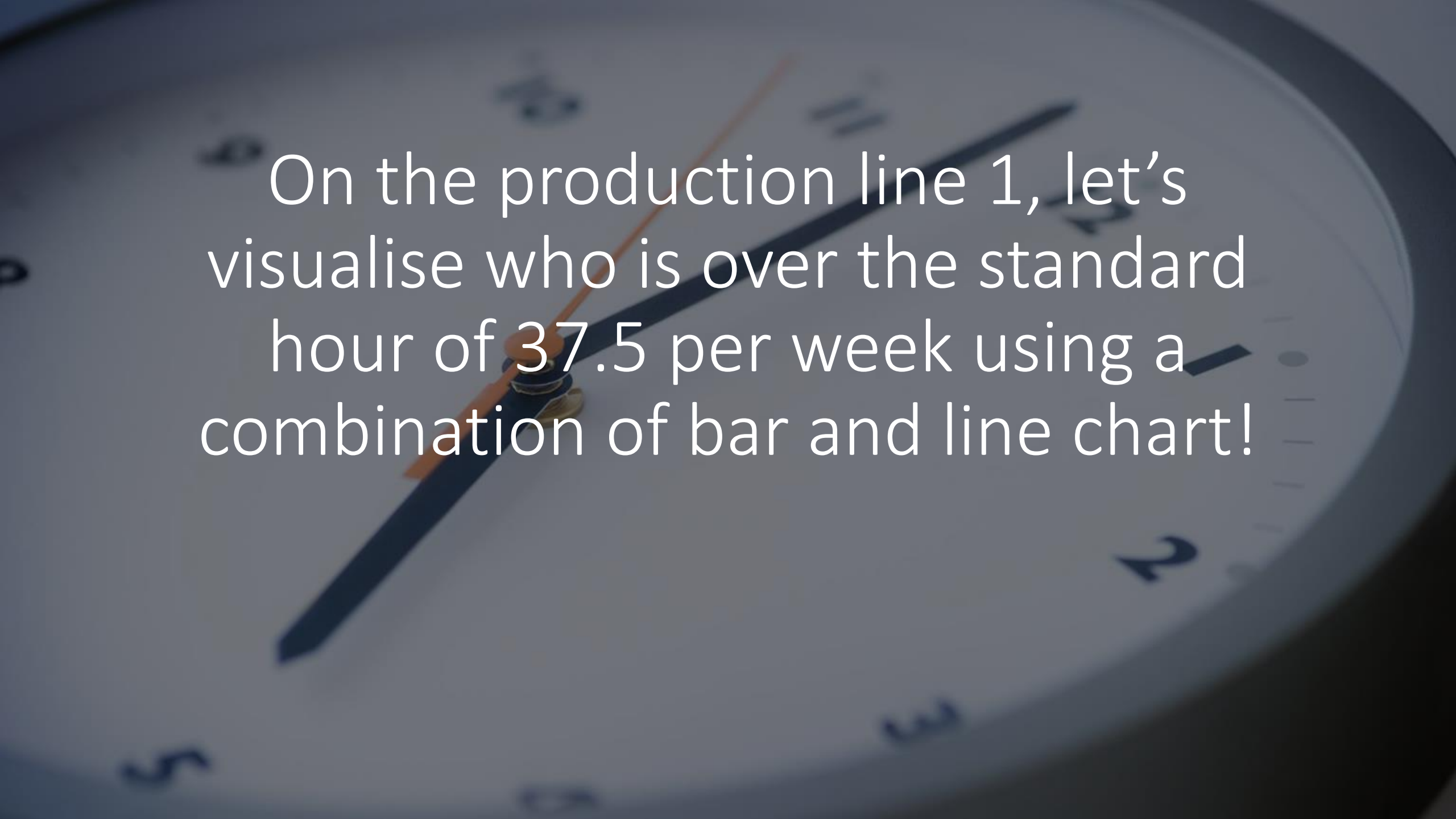


Show how a value changes over time, though the independent variable can really be anything. Since they are most commonly used to visually represent trends over time, **line charts are sometimes referred to as time-series charts.**

- **Advantages:** show trends and relationships between data better than other graphs; compare trends in different groups of a variable; clearly show error values in the data; Usually simple to read and understand

Median dwelling values





On the production line 1, let's visualise who is over the standard hour of 37.5 per week using a combination of bar and line chart!

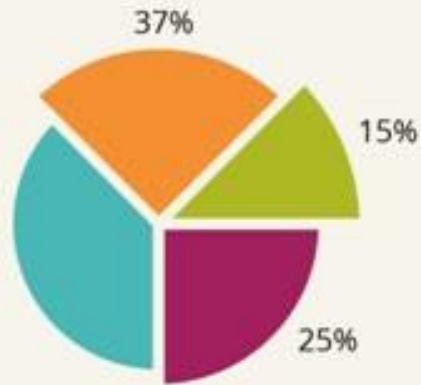
Pie Chart



Pie Graph



Pie Graph

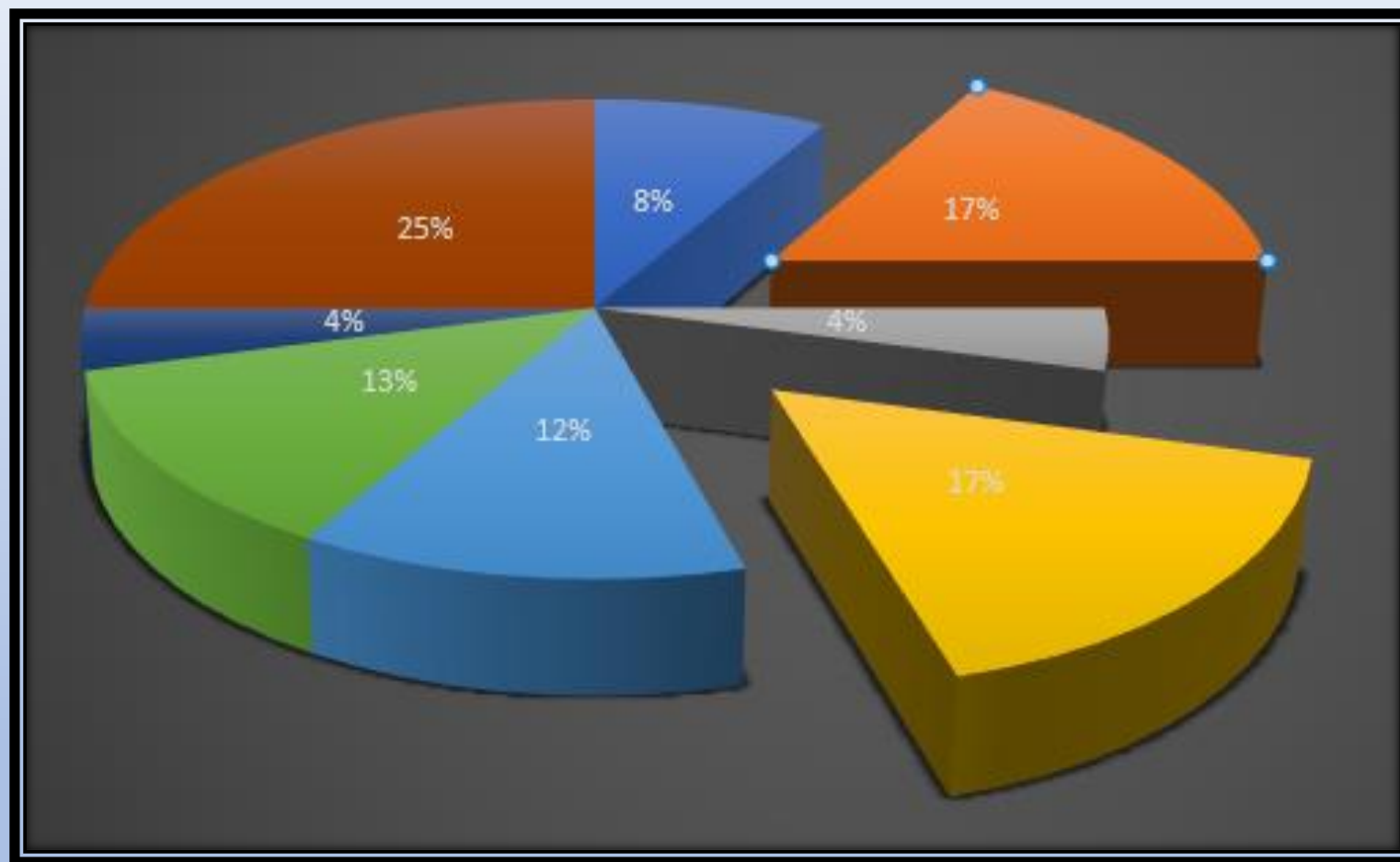


Typically used to summarise categorical data, or mostly percentile value.

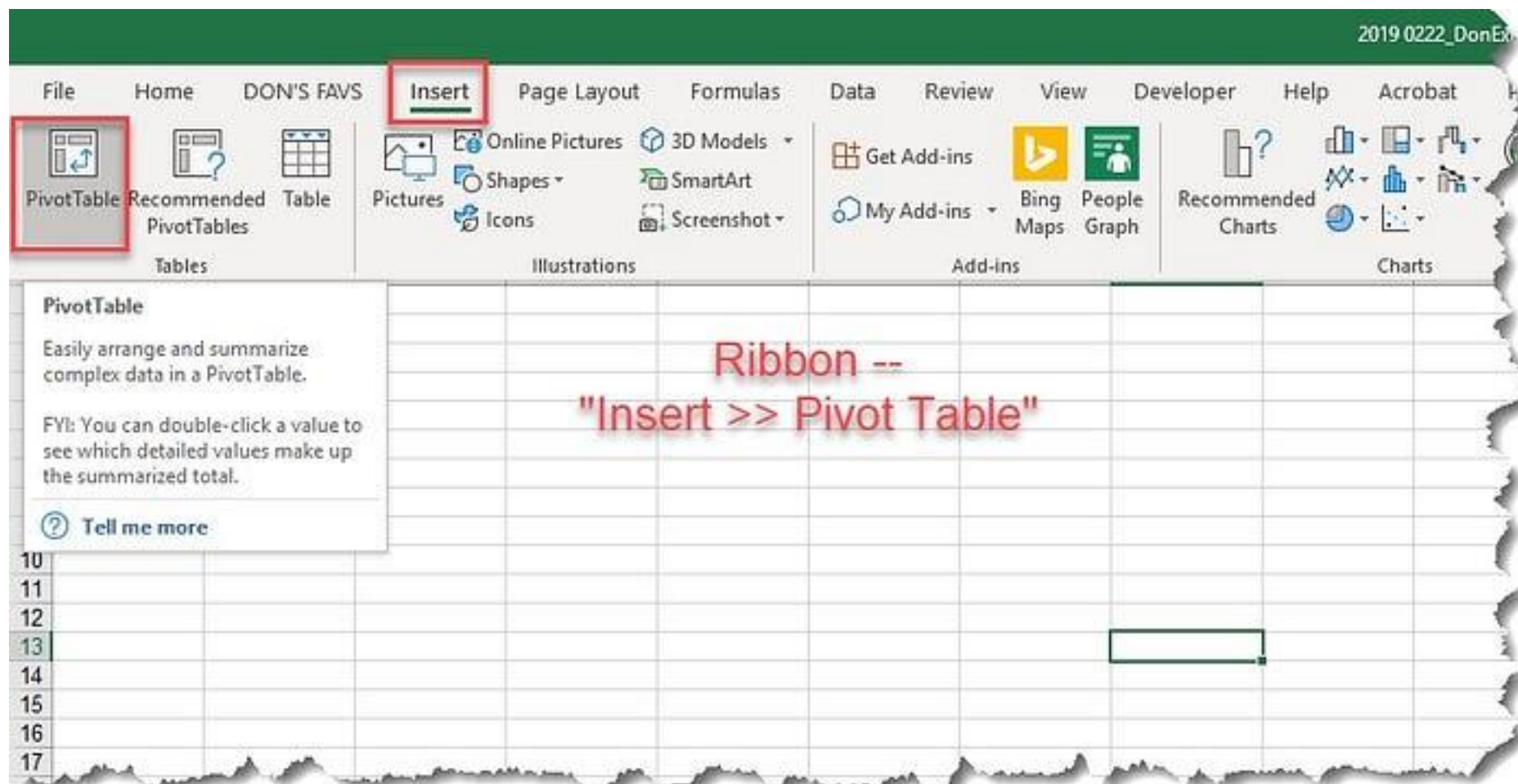
➤ **Advantages:**

- ❑ It provides an excellent visual concept of a whole; clear comparison of different components,
- ❑ It exposes information by the visual separation of a segment
- ❑ It is easy to label.

➤ **Variations:** Donut Graph



Pivot tables



It is particularly useful when **one of the variables** (represented by the y-axis) **is dependent on another** (represented by the x-axis). The resulting pattern (after all the points have been plotted) will indicate the strength of the correlation between two variables.

- **Advantages:** Clearly indicates **data correlation** (illustrates

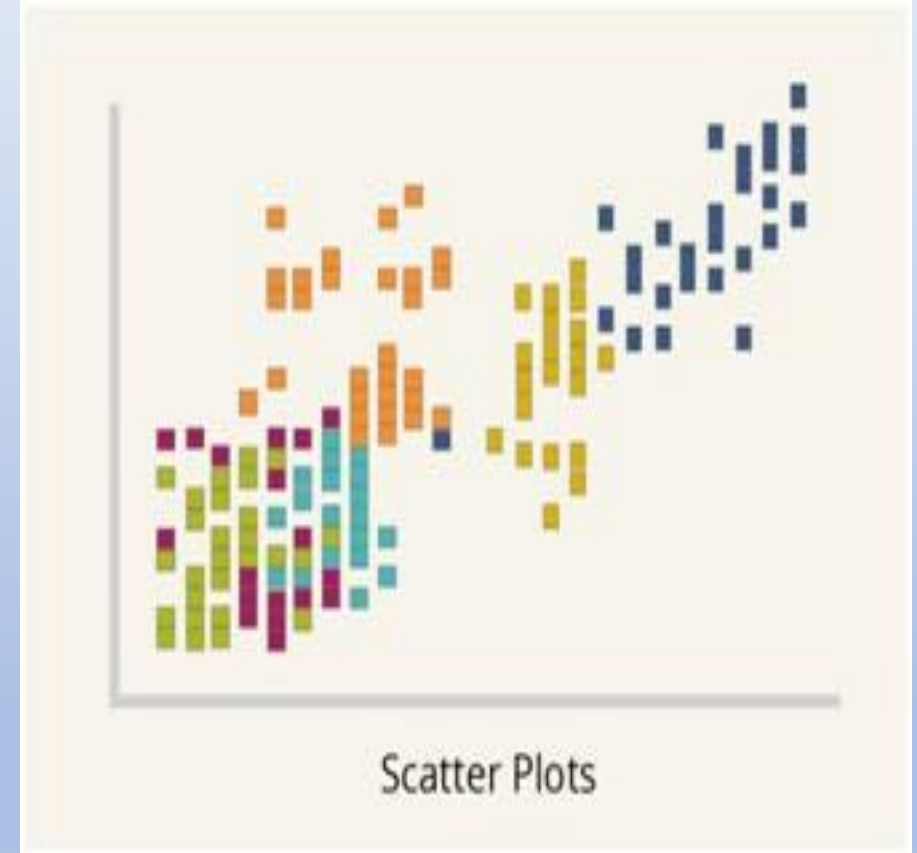
positive, negative, strong, and weak correlations);
method of

illustration **non-linear patterns**; shows spread of data,
outliers;

clearly demonstrate atypical relationships; used for data
extrapolation and interpolation

- **Disadvantages:** **impossible to label data points**,
hard to find out exact values; error bars and too many data
points can quickly make graph unreadable; **cannot show
relationship between more than two variables at once.**

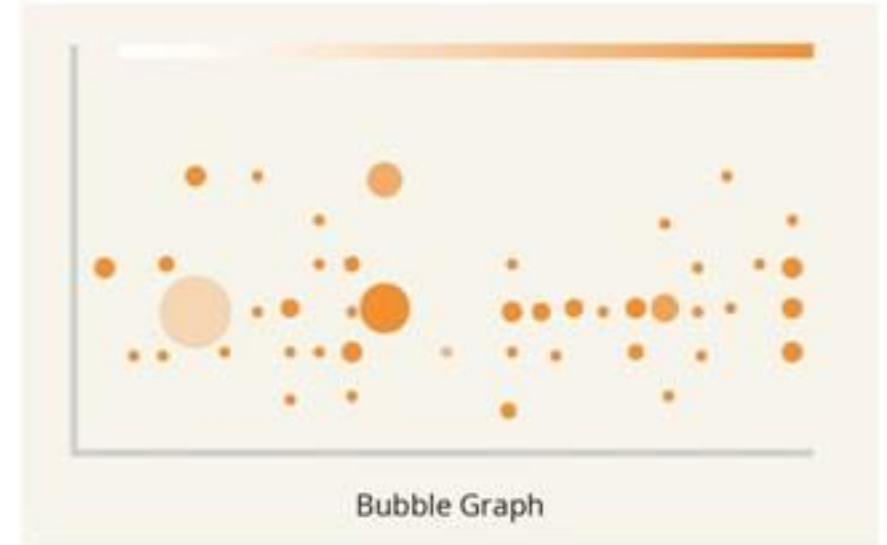
Scatter plots



Bubble graphs are useful for comparing the relationships between data objects in 3 numeric-data dimensions: the x-axis data, the y-axis data, and data represented by the bubble size. Bubble charts are often used in business to visualise the relationships between alternatives investment in dimensions such as cost, value, and risk.

- **Advantages:** display **three variables without using 3D graphs**; visual size makes it very **easy to make relative comparisons**; Conveys same information as a line graph
- **Disadvantages:** due to circle sizes, can be difficult to ascertain actual values; difficult to read and understand; **cannot be used to display a lot of data.**

Bubble graph



A Radar charts are very useful when comparing performance/measurement **results from different sources**. Obviously, the greater the area covered by the plot, the greater the overall value. Through comparison and analysis, people can figure out their current situation: improving or falling behind.

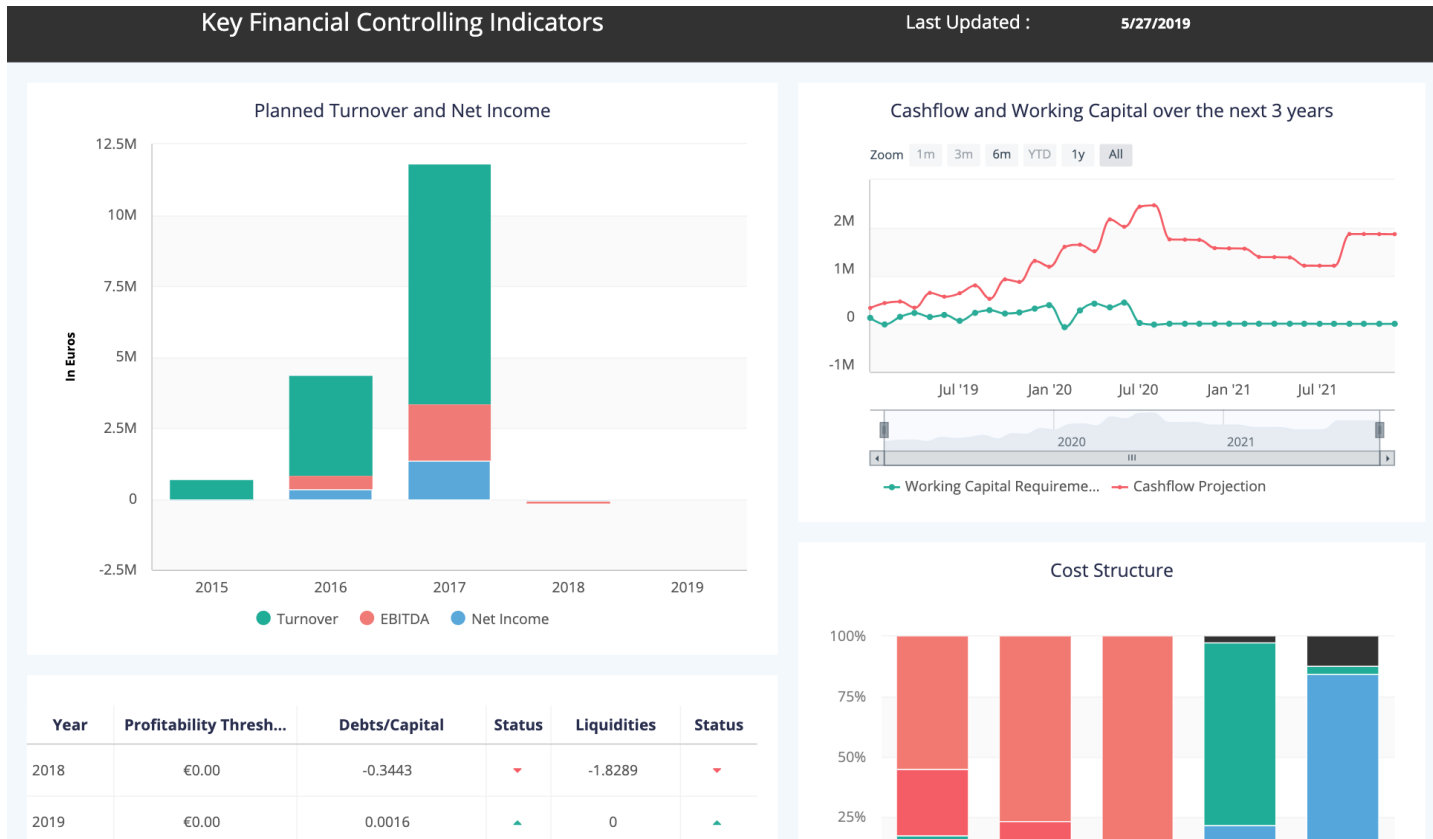
- **Advantages:** primary way of displaying **more than two or three values at once**; excellent way to get a "feel" for data;
- **Disadvantages:** cannot compare more than two or three different plots at once; without colouring, can be difficult to tell which points belong to which; **too many axis makes it difficult to read** less intuitive than other graph types

Radar Chart (Spider chart)



Radar Chart (Spider Chart)

Dashboarding: Chart vs. Dashboard



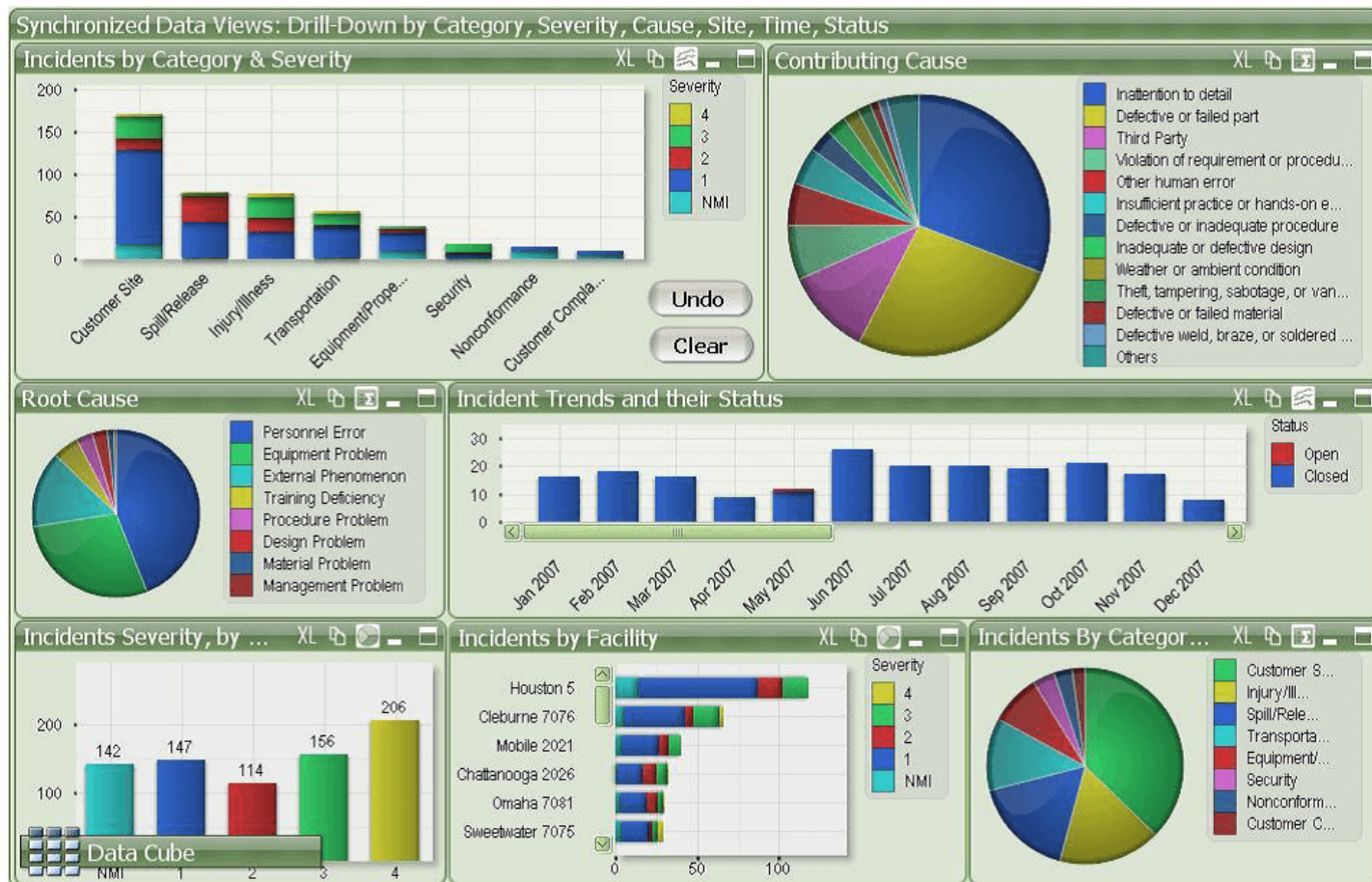
A chart or a graph:

- A **chart**, or a graph, is the display of a specific information (**a chapter of a story**).

A dashboard:

- A dashboard is a **collection of these** (chapters come together to tell a story)
- Your dashboard **needs to narrate a story**; it needs to have a goal, and some objectives.
- It is arranged on a single screen so the information can be monitored at a glance

Dashboarding

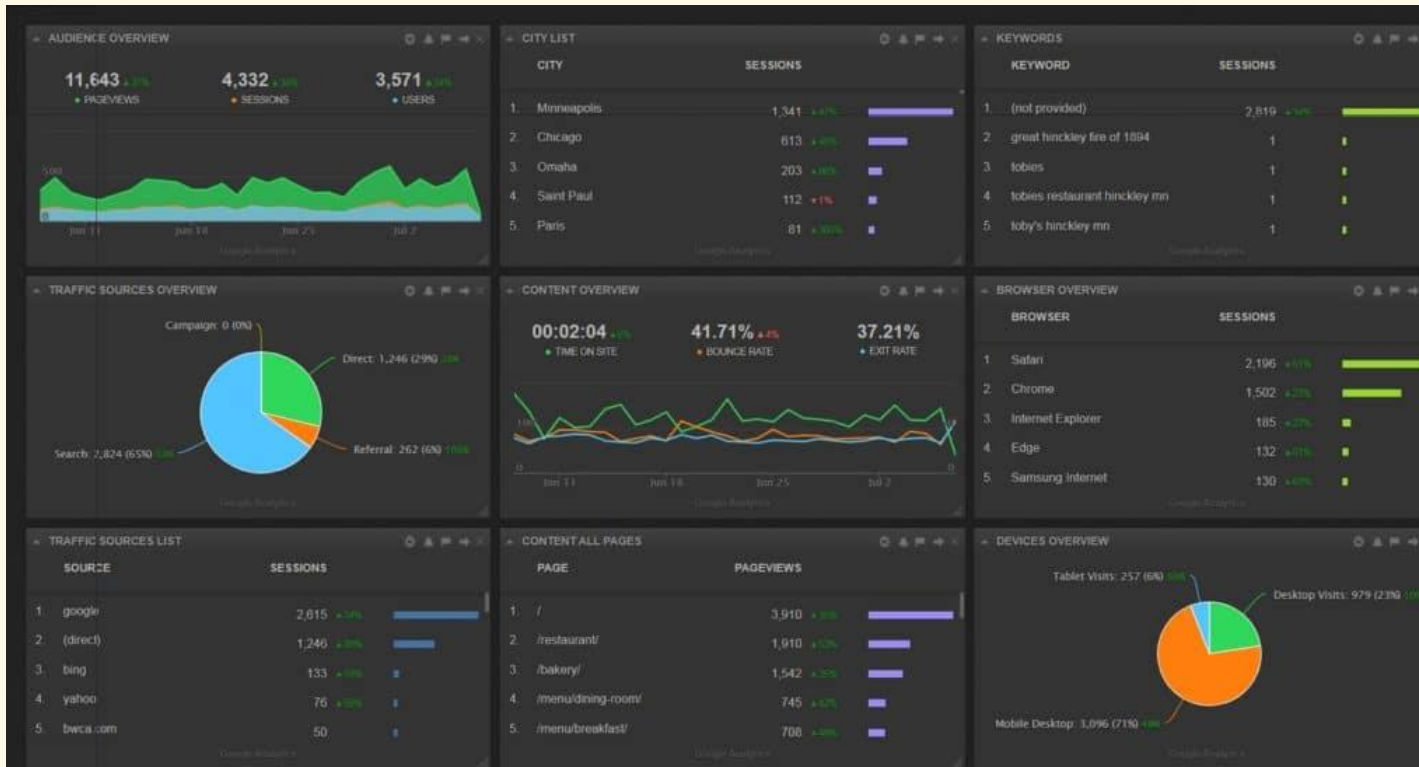


A dashboard:

- Usually brings data from different data sets, or different part of a big data together to visualise them and tell a story.
- Data that are brought together need to be **related and connected**
- Where you have, include the display **Key Performance Indicators (KPIs)**.

Note: **All charts in Excel change as you change the source of data.** This helps your dashboard to be updated as data is updated.

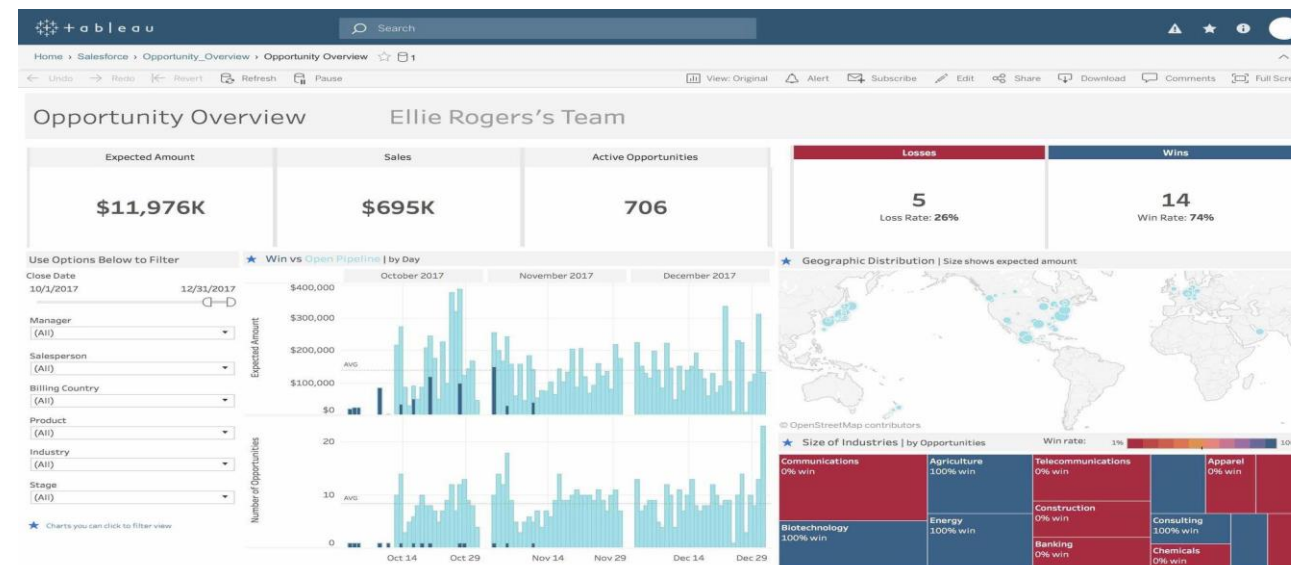
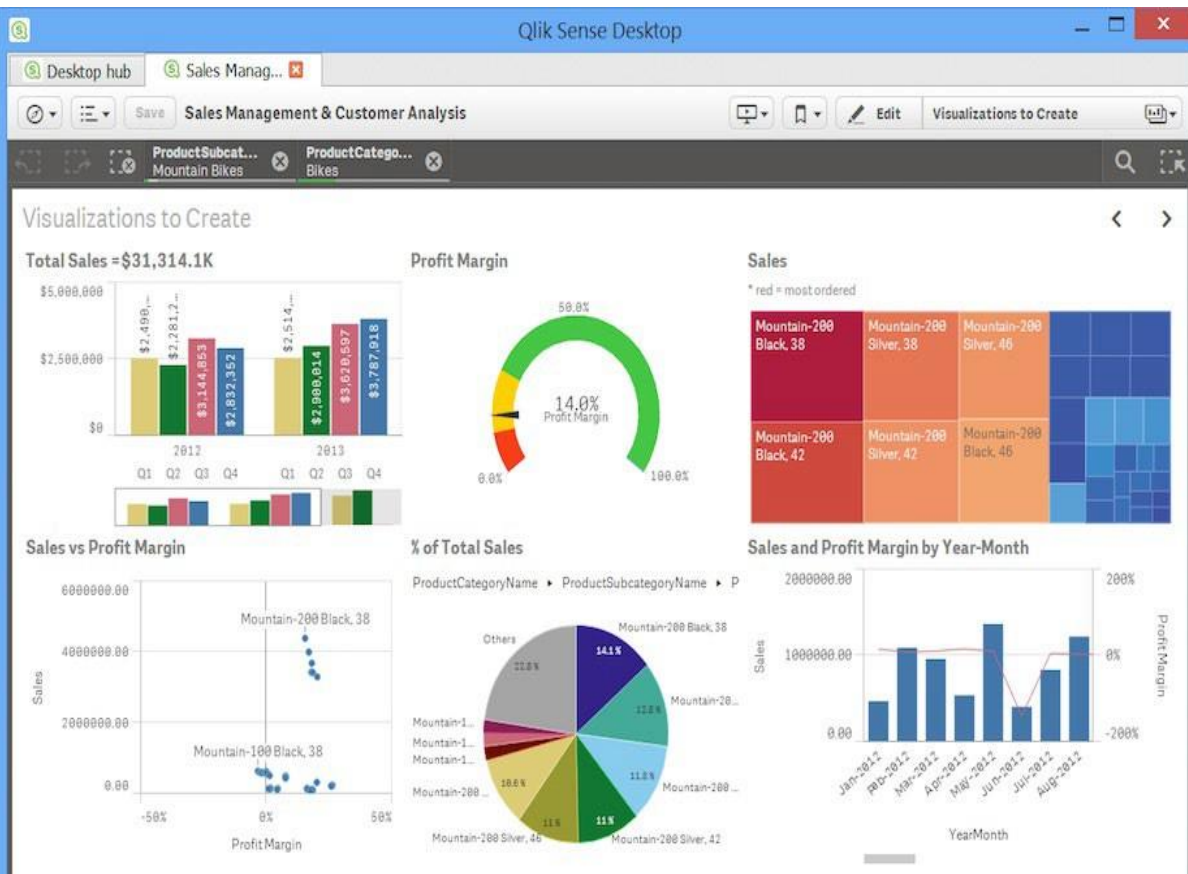
Characteristics of Dashboards



- Customisable
- Interactive
- Real-Time monitoring
- All of the data is in one place
- Get everyone on the same page
- Attractive (capable of gaining viewers focus)

Other Data visualisation & Dashboarding tools

- PowerBI
- QlikView
- Tableau



Dashboard in Excel

- It is an enhanced form of Data Visualisation.
- A data source for creating Excel dashboards can be Spreadsheets, Text Files, Web Pages, Organizational Database.

Examples:

<https://gisanddata.maps.arcgis.com/apps/opsdashboard/index.html#/bda7594740fd40299423467b48e9ecf6>

<http://www.cidrap.umn.edu/covid-19/maps-visuals>

(M/D/YYYY)
5:20 PM

Total Cases

480,115,822

Total Deaths

6,121,156

Total Vaccine Doses Administered

10,868,565,572

28-Day Cases

45,462,311

28-Day Deaths

181,049

28-Day Vaccine Doses Administered

392,836,869

aths by
/Sovereignty

00 | 6,955
| 14,899

80 | 2,208
42,258

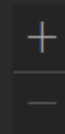
50 | 4,907
| 127,599

82 | 3,591
| 142,664

67 | 3,249
| 165,046

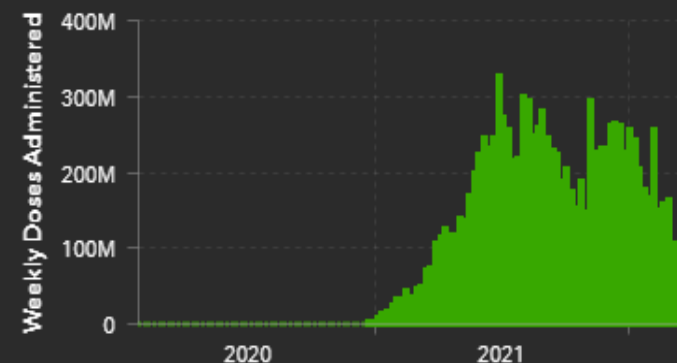
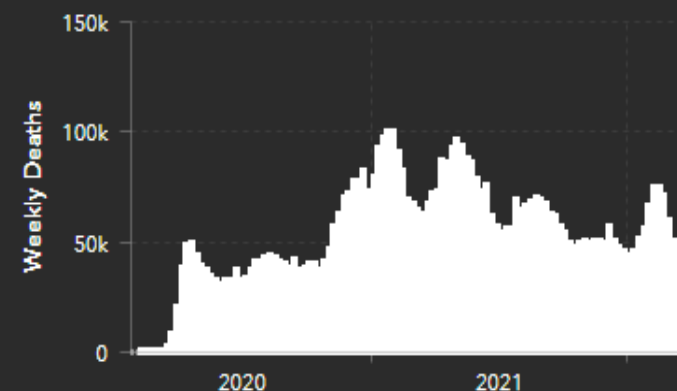
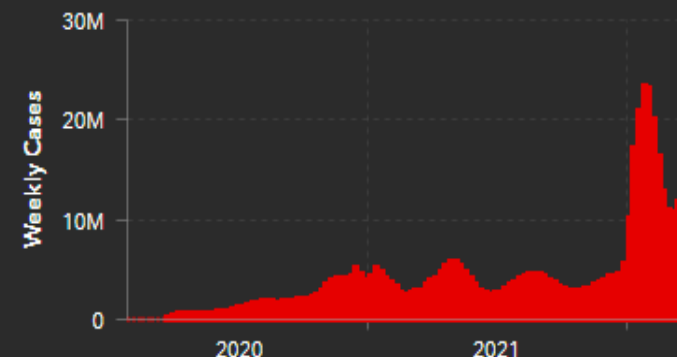
31 | 4,284
| 158,700

58 | 16,515
| 359,693



Esri, FAO, NOAA

Powered by Esri



n0 ▶

28-Day

Totals

Incidence

Case-Fatality Ratio

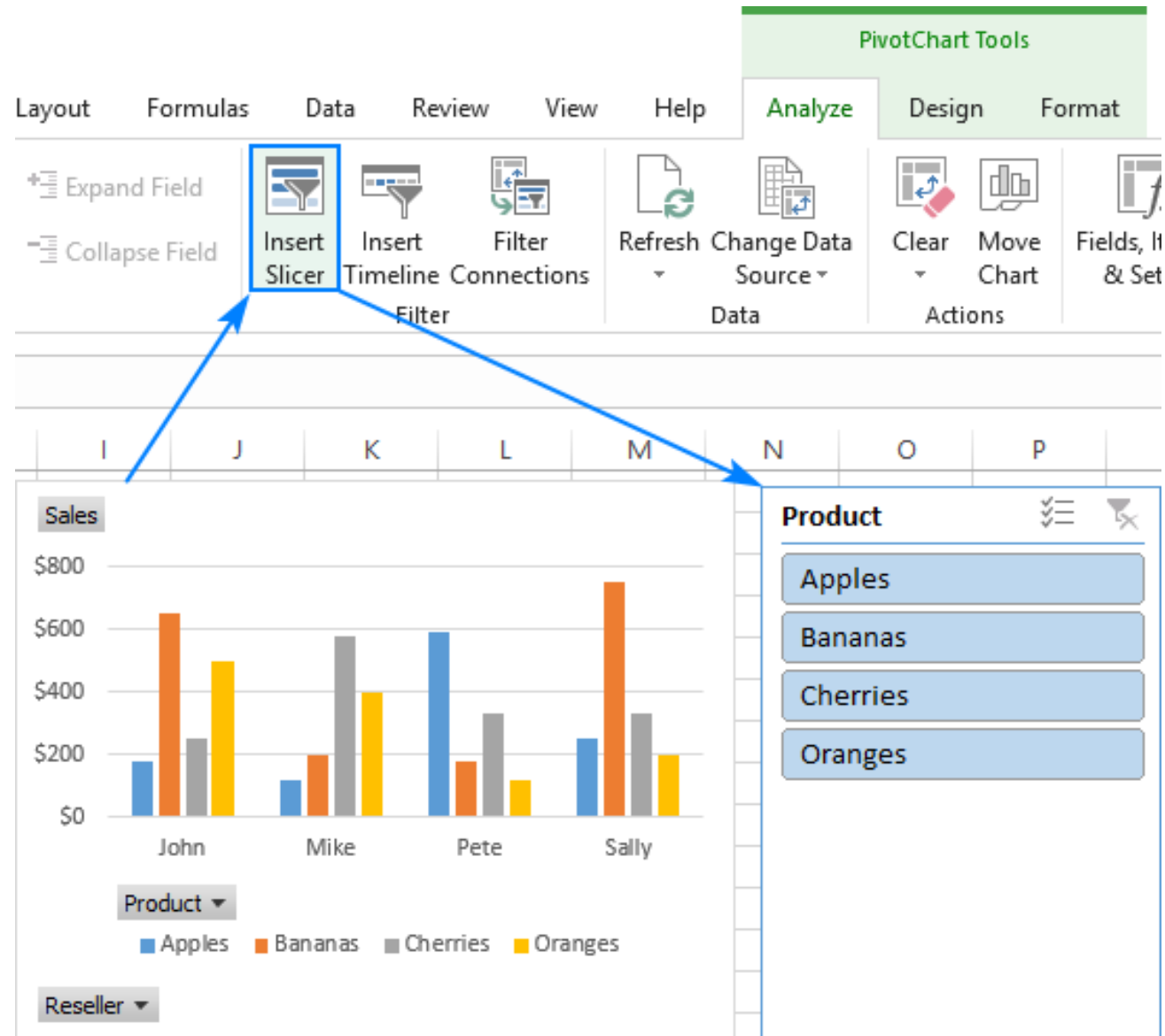
Global Vaccinations

US Vaccinations

Terms of Use

Weekly

28-Day





An Effective Dashboard May Include Key Performance Indicators (KPIs) to facilitate Efficient Monitoring of the Performance of your Supply Chain.



The Importance of Monitoring Supply Chain Performance

Cost Reduction: By identifying areas of waste and inefficiency, organisations can reduce costs.

Customer Satisfaction: Ensuring availability of inventory, on-time delivery and product quality

Risk Mitigation: Measuring performance helps identify and mitigate risks.

Supplier Relationships: Effective performance measurement enhances communication and collaboration with suppliers and enhances the chance of establishing long-term relationships with them.